



Energy Consumption Optimizer

SLIDE-BY-SLIDE EXPLANATORY NOTES

COURSE: MACHINE LEARNING LAB (SEM-2)

Case Study | Regression | Feature Scaling | Model Deployment

Focus: Explaining pipeline steps, data engineering decisions, and evaluation results.

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HOW TO USE THIS GUIDE

This document provides slide-by-slide notes designed to guide you through your viva or class presentation. To deliver a high-scoring ML Lab presentation, do not just read the text on the slides. Instead, explain the logic behind the steps, call out engineering and math principles, and use the transition cues to maintain a smooth flow.

Each slide section below covers:

1. Slide Objective: The core idea you want the listener to grasp.
2. Key Talking Points: What you should speak while displaying the slide.
3. Technical Context: Background math, code logic, or metrics you can highlight to show expertise.
4. Transition Cue: Suggested verbal transitions to the next slide.

Slide 1. Title & Cover Slide

SLIDE OBJECTIVE

Establish a professional initial impression and introduce the team, task, and case study focus.

KEY TALKING POINTS (WHAT TO SAY)

* **Introduce the Team:** Good morning/afternoon, we are Gaurav, Akash, and Mithun. Today we are presenting our Semester 2 Machine Learning Lab case study: the Energy Consumption Optimizer.

* **State the ML Task:** This is an end-to-end regression task centered on hourly electricity load prediction for smart cities. Our primary focus is on Feature Scaling techniques and a web-based Deployment framework.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Mention that this is a classic regression case study where the objective is to predict a continuous target variable (electricity load in kWh) using calendar and meteorological features.

TRANSITION CUE

Let's begin by discussing what this project is about and the system goals.

Slide 2. Project Overview (Introduction)

SLIDE OBJECTIVE

Explain the business/domain objective of the smart city load prediction.

KEY TALKING POINTS (WHAT TO SAY)

- * **Grid Balance:** Explain that power grids cannot store massive quantities of electricity. Production must match demand dynamically. Overproduction creates economic and environmental waste, whereas underproduction leads to blackouts.

- * **ML Objective:** Our objective is to forecast electricity load a few hours ahead of time using readily available grid signals: time of day, day of week, month, temperature, and humidity.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Specify the input variables (Hour: 0-23, Day of Week: 0-6, Month: 1-12, Temperature: degrees C, Humidity: %) and the target variable (electricity_load: kWh). Highlight that weather variables directly drive HVAC usage, which represents over 40% of grid load.

TRANSITION CUE

While this sounds straightforward, working with raw sensor data introduces immediate challenges. Let's look at the data problems we had to solve.

Slide 3. Real-world Data Challenges (Problem Statement)

SLIDE OBJECTIVE

Highlight the common anomalies present in raw sensor telemetry and define why pre-processing is mandatory.

KEY TALKING POINTS (WHAT TO SAY)

- * **Telemetry Anomalies:** Point out that IoT sensors are prone to failures. They lose network packets leading to missing values, or record duplicate logs due to transmission re-tries.
- * **Scale Imbalance:** Features naturally exist on different scales. Humidity values (40-90) are numerically larger than hours (0-23) or temperatures. A regression model will wrongly assume that larger numbers are more important.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Quantify the anomalies found in the raw data: 140 missing sensor inputs (80 temperature, 60 humidity) and 30 duplicate records. Explain that unscaled inputs cause gradient descent to oscillate or distance metrics to skew towards high-magnitude features.

TRANSITION CUE

To solve these problems, we built a structured, step-by-step pipeline.

Slide 4. Machine Learning Pipeline Workflow

SLIDE OBJECTIVE

Give the audience a high-level visual roadmap of the project's engineering stages.

KEY TALKING POINTS (WHAT TO SAY)

- * **The 6 Stages:** Walk through the 6 stages of our project: (1) Synthetic Data Simulation, (2) Data Cleaning, (3) Exploratory Data Analysis, (4) Feature Scaling, (5) Model Training, and (6) API Deployment.

- * **Pipeline Integrity:** Emphasize that a machine learning project is not just training a model; 80% of the work lies in data preparation and deployment preparation.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Mention that maintaining a modular pipeline allows us to easily swap models (e.g. going from Linear Regression to Random Forest) or scale up the dataset without rewriting the preprocessing steps.

TRANSITION CUE

Let's dive into the details of the first two stages: data simulation and cleaning.

Slide 5. Dataset Simulation & Preprocessing

SLIDE OBJECTIVE

Detail how the dataset was established and how data anomalies were handled.

KEY TALKING POINTS (WHAT TO SAY)

- * **Data Generation:** To model a full year of hourly logs, we simulated 8,760 hours of records. The simulation uses sine waves to model diurnal temperature and load cycles, and applies a weekend drop of 30 kWh to mimic closed business offices.
- * **Data Preprocessing:** We resolved duplicate logs by dropping them. We solved the missing values by imputing them with the average value of each respective column.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Mention that dropping duplicates reduces model bias. For missing values, mean imputation (`df.fillna(df.mean())`) was selected. In production, we must use the training mean for imputation on incoming live data to prevent leakage.

TRANSITION CUE

With clean data in hand, we explored the patterns visually.

Slide 6. Exploratory Data Analysis (EDA) - Load Patterns

SLIDE OBJECTIVE

Explain the major cyclical patterns observed in the electricity load.

KEY TALKING POINTS (WHAT TO SAY)

- * **Daily Peak:** Point out that electricity load shows a distinct diurnal pattern. It is lowest between 3 AM and 5 AM when the city sleeps, rises as the day starts, peaks in the afternoon/evening, and drops again at night.
- * **Weekly Variance:** Point out that weekends show a clear baseline drop of approximately 30 kWh compared to weekdays, which reflects reduced commercial activity.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Explain that grouping the data by hour and plotting averages reveals a classic bimodal load curve. Refer to the plot in the slide showing the actual 7-day cyclical waveforms generated in our EDA notebook.

TRANSITION CUE

Next, let's look at the correlation heatmap to quantify the relationship between features and the load.

Slide 7. What Drives Electricity Load? (Correlation Analysis)

SLIDE OBJECTIVE

Quantify relationships between variables using Pearson Correlation Coefficients.

KEY TALKING POINTS (WHAT TO SAY)

- * **Temperature Factor:** Highlight that temperature is positively correlated with load. As temperature goes up, load goes up, capturing air conditioning usage on hot summer days.
- * **Weekend Drop:** Show the negative correlation for weekends, confirming our baseline drop of 30 kWh on Saturdays and Sundays.
- * **Hour Correlation:** Mention that the hour of the day has a positive correlation, reflecting the rising active demand during the day.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Define correlation range (-1.0 to +1.0). Explain that while correlation measures linear relationships, some features (like temperature) have non-linear relationships with load (extreme cold also increases heating demand, creating a V-shape).

TRANSITION CUE

Knowing how these features behave, we now face the challenge of different scales. Let's look at how we resolved this.

Slide 8. Feature Scaling Deep Dive

SLIDE OBJECTIVE

Explain the mathematical differences between scaling techniques and how they prevent data leakage.

KEY TALKING POINTS (WHAT TO SAY)

- * **The Scaling Problem:** Recall that hour (0-23) and temperature (5-45) are small, while humidity (40-90) is large. A model like Linear Regression will assign larger weight updates to humidity, slowing down training or skewing predictions.

- * **Our Scalers:** We evaluated StandardScaler, which adjusts values to mean=0 and std=1, and MinMaxScaler, which squashes values into the 0-1 range.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Highlight the Scaling Rule: always call `fit_transform()` on train data to learn the mean and variance parameters, and call `transform()` on test/live data. Fitting on test data causes data leakage, violating evaluation integrity.

TRANSITION CUE

Lab Note: Scaling Rule - Never fit your scaler on test data! Always call `fit_transform()` on the training set, and only `transform()` on the test set to avoid data leakage.

Slide 9. Linear Regression vs. Random Forest (Model Comparison)

SLIDE OBJECTIVE

Show performance metrics and explain why the non-linear model vastly outperformed the linear model.

KEY TALKING POINTS (WHAT TO SAY)

- * **Linear Regression:** Linear Regression scored an R^2 of only 1.92%. This means the model failed to capture the electricity load curves. A straight line cannot represent a daily sine wave or a V-shaped temperature effect.
- * **Random Forest:** Random Forest achieved an R^2 of 91.12%. It uses 100 decision trees to segment the feature space, easily modeling non-linear waves and absolute temperature boundaries.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Present the exact metrics: Linear Regression MSE is 5424.45 (RMSE: 73.65 kWh); Random Forest MSE is 491.14 (RMSE: 22.16 kWh). Explain that MSE penalizes large errors. R^2 measures the proportion of variance explained by the model.

TRANSITION CUE

To show how this works in real-time, we built a live, interactive simulator on our dashboard.

Slide 10. Interactive Grid Load Predictor

SLIDE OBJECTIVE

Demonstrate the models making predictions in real-time under different grid conditions.

KEY TALKING POINTS (WHAT TO SAY)

- * **Interactive Demo:** Direct the audience to the slide's controls. We can adjust hour, day of the week, month, temperature, and humidity, and see both model predictions update in real-time.

- * **Key Behaviors:** Note that if we slide the temperature to a hot summer day like 35 degrees C at 6 PM, the predicted load peaks. On a weekend, the load drops as expected.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Explain that the Linear Regression prediction uses the exact coefficients (hour: +6.71, is_weekend: -14.11, etc.) and scaler parameters (means and scales) dumped from your python model file: scaling is done on the fly in JS.

TRANSITION CUE

Now that we have a highly accurate model, how do we deploy this to the actual smart city grid?

Slide 11. API & Production Architecture

SLIDE OBJECTIVE

Describe the production deployment pipeline for real-time inference.

KEY TALKING POINTS (WHAT TO SAY)

- * **Saving Models:** We serialize both the trained model and standard scaler parameters into bin files using Joblib. This preserves the weights and prevents retraining.
- * **Web Service:** We wrap the files in a web application (using Flask or FastAPI). The service exposes a POST API endpoint that accepts grid readings, scales them, runs predictions, and returns a JSON payload.
- * **Cloud Integration:** This application is hosted on AWS (EC2/SageMaker). The city operators' dashboard calls this API hourly to adjust energy production levels.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Show the JSON input/output payload structure. Point out that the scaler.pkl must be deployed along with the model.pkl, since incoming live data is unscaled and must be transformed using the training parameters.

TRANSITION CUE

Let's summarize our findings and key takeaways from this project.

Slide 12. Conclusion & Takeaways

SLIDE OBJECTIVE

Provide a summary of the project accomplishments and key engineering lessons.

KEY TALKING POINTS (WHAT TO SAY)

- * **Summary of Work:** We simulated, cleaned, scaled, trained, and saved a complete machine learning pipeline for smart city electricity optimization.

- * **Lessons Learned:** We learned that data cleaning is vital, feature scaling prevents model scale bias, and non-linear patterns (diurnal waves) require non-linear models like Random Forests.

TECHNICAL CONTEXT (SHOW EXPERTISE)

Reinforce that model selection must align with data patterns. Linear regression is insufficient for complex cyclical loads. Future enhancements could include using LSTM models for sequential time-series forecasting.

TRANSITION CUE

Thank you. We are now open to any questions you may have.